

Politician Trade Agent

An **Unusual Whales-class** market terminal — congressional trade tracking, real-time options flow, and prediction-market wallet forensics — rebuilt entirely on **free public data** and piped to Discord, a web dashboard, and your phone.



Congress filings



Real-time options



Polymarket forensics



≈ \$5 / month



What this is.

A one-page tour of the idea plus the exact steps to take it live. The pitch is up top; the phased runbook (\$4) is what you follow tomorrow to go from a fresh clone to a running live test — every step has a pass/fail gate, and nothing here costs more than a coffee.



The idea in one breath

Commercial tools like Unusual Whales charge a monthly subscription to watch **where the informed money moves** — Congress, the options tape, dark pools. Almost all of that signal is sitting in **free, public feeds**: House & Senate disclosure filings, a free brokerage API for real-time options quotes, FINRA aggregates, and Polymarket's open trade feed. This project stitches them together and pushes the alerts to you — for the price of a tiny VPS.

public data sources



parse & enrich



detect the unusual



Discord · dashboard · phone



What it does



Congressional trading. Polls official House Clerk & Senate EFD filings and parses **per-trade detail** — ticker, buy/sell, dates, amount range, owner. Posts alerts with party/state tags and disclosure-lag, plus profiles ([/politician](#)), most-traded tickers ([/top](#)), and a return-vs-SPY leaderboard ([/leaderboard](#)). Optional: mirror the trades through Alpaca (paper or live).



Options flow & market data. [/flow](#) for unusual contracts & put/call sentiment, [/gex](#) for Black-Scholes gamma exposure, [/darkpool](#) for FINRA short-volume & ATS data, plus background unusual-activity alerts. Drop in a **free Tradier key** and every consumer upgrades to **real-time OPRA quotes** automatically — no key means a delayed Yahoo fallback.



Prediction-market forensics (Polymarket). Watches the public trade feed and profiles the wallet behind every large bet — **fresh wallets** suddenly placing size, **sharp wallets** with outlier win records, **longshot conviction**, and **whale-size** fills. Something Unusual Whales has no equivalent for.

🕸 **Insider-surge detection & wallet clustering.** The 60 Minutes / Bubblemaps pattern: **many wallets converging one-sided** on the same outcome before a strike. Category-weighted (military & geopolitical score highest), with behavioral co-bet clustering always on and an optional on-chain funder-graph via a free Polygonscan key. [/surges](#) lists the hits.

📱 **Web dashboard & phone app.** A responsive dashboard with live feeds and per-ticker tools, backed by JSON APIs — installable as a **PWA** (Add to Home Screen) for a standalone phone app. Token auth gates it; [/api/health](#) feeds any uptime monitor.

vs. Unusual Whales.

Congressional tracker, options flow, Greeks/GEX, dark pool, and Discord alerts — all matched on free data. Then it goes **further**: trade execution (Alpaca), prediction-market wallet forensics, and Bubblemaps-style surge detection, none of which UW offers.

🔌 Data sources & plumbing

Signal	Source	Cost
Congressional trades	House Clerk XML + Senate EFD (parsed PTRs)	Free
Options chains / Greeks	Tradier real-time (free key) · Yahoo delayed fallback	Free
Dark pool / short volume	FINRA daily & weekly aggregates	Free
Prediction markets	Polymarket public data API	Free
Wallet funder graph	Polygonscan (optional free key)	Free
Politician enrichment	unitedstates/congress-legislators dataset	Free
Trade execution	Alpaca paper / live	\$0 commissions

SQLite storage with full dedup (no duplicate alerts or orders); the first run backfills history silently instead of flooding the channel. HTTP retries with backoff and an ETag-aware cache keep it polite to government servers.

4 Live setup — the path for tomorrow

🕸 **The golden rule.**

Every integration was built and unit-tested, but the government/market APIs couldn't be reached from the build sandbox. So the phases below are ordered to **prove each data source before anything depends on it**. Every source degrades to a warning, never a crash — when a step fails, capture the log and move on.

0

Clone & install — Python 3.11+, no paid services

```

bash — bootstrap

git clone <your-repo-url> && cd politician-trade-agent
git checkout main
python -m venv .venv && source .venv/bin/activate
pip install -e ".[dev,trading]"
pytest -q          # expect: all tests pass before you start
cp .env.example .env

```

1

Congressional smoke test — no credentials needed

The single most important check: exercises the House ZIP + PDF parsing, the Senate EFD handshake + HTML parsing, and enrichment — no Discord, no broker.

```

bash

python -m src.main --once

```

PASS `--once complete: fetched=N new=N` with $N > 0$, and some **NEW** lines show **real tickers & amount ranges** (proves detail parsing). Run it twice — the second run should show `new=0` (dedup proof).

2

Market-data smoke test — still no credentials

A short Python snippet (in the runbook) hits Yahoo options, FINRA short volume, and the Polymarket feed. **PASS** = all three print **OK**. A Yahoo miss only affects the keyless fallback — Tradier (\$Phase 5) makes it moot.

3

Discord bot — token + channel ID

Create the app in the Developer Portal, invite with scopes `bot` + `applications.commands` (Send Messages, Embed Links, Use Slash Commands), and copy your alerts channel ID into `.env`.

PASS Typing `/` lists every command — `trades · top · politician · leaderboard · follow · flow · gex · darkpool · polymarket · surges · portfolio · settings`. The first cycle logs a **silent backfill** and posts nothing, by design.

4

Dashboard + phone app — optional, 2 minutes

```

.env

ENABLE_DASHBOARD=true
DASHBOARD_AUTH_TOKEN=$(python -c "import secrets;print(secrets.token_urlsafe(24))")

```

From your phone on the same Wi-Fi: `http://<lan-ip>:8080/?token=...`, then browser menu → **Add to Home Screen**. **PASS** = 401 without the token, loads with it, panels populate, and `/api/health` shows a fresh `last_poll_age_s`.

5

Real-time options data — optional, recommended, \$0

Open a free Tradier brokerage account (no funding needed for API access), create a **production** token, set `TRADIER_API_KEY`. That's it — `OPTIONS_PROVIDER=auto` upgrades flow, `/flow`, `/gex`, and the dashboard to real-time. Verify `/gex SPY` during market hours; spot should match a live quote to the penny.

6

Scanners — flow, prediction markets & surges

```
.env

ENABLE_FLOW_ALERTS=true
FLOW_MIN_PREMIUM_USD=250000
ENABLE_PREDICTION_ALERTS=true
PREDICTION_MIN_BET_USD=10000
ENABLE_SURGE_ALERTS=true # surges run inside the prediction scanner
```

Give it a full trading day. **Too noisy?** raise the premium / bet floors and `SURGE_MIN_WALLETS`. **Too quiet?** lower them. Expect one tuning pass after day one and another after a week.

7

Paper trading → **always-on** → **live** — take your time

Turn on Alpaca **paper** trading (buys only) and run it for at least two clean weeks — orders appear in Alpaca, restarts never duplicate, `/portfolio` matches. Then move it to an always-on host (VPS + systemd, or Docker), and only after weeks of clean paper flip to **live** keys — small size, buys only, checked daily.

🚦 Before you rely on it

**Strategy reality.**

The STOCK Act allows members up to **45 days** to disclose, so you always enter late and amounts are ranges, not exact sizes. Treat mirroring as one signal input, not arbitrage.

**Data licensing gate.**

Tradier real-time quotes are licensed for **your personal use**. Don't expose a public dashboard with a Tradier key attached — keep it token-protected and private, or run keyless (delayed) if you must share it.

**Source fragility.**

house.gov / senate.gov change markup without notice; Yahoo & Polymarket endpoints are unofficial. Watch `/api/health` and the logs weekly; budget the occasional parser fix.

💰 What it costs

All data sources (Congress, FINRA, Polymarket, enrichment)	\$0
Tradier — real-time options data	\$0
Alpaca — paper & live trading	\$0 commissions
Discord	\$0
VPS hosting (only once you go always-on)	~\$4–12 / mo
Total to run it live	≈ \$5 / month

⚡ TL;DR

A subscription-grade market terminal — Congress, options flow, dark pool, prediction-market forensics, alerts, a phone app, and optional auto-trading — built on free public data for the price of a small VPS. Clone, run `--once` to prove the data, wire up Discord, then layer in the dashboard, real-time quotes, and scanners. Paper-trade for weeks before real money.

🔴 🟡 🟢 quickstart

```
python -m src.main --once # prove the pipeline, no credentials
python -m src.main        # then go live with Discord
```